

Project Design Phase-I Solution Architecture

Date	19 June 2026
Team ID	-----
Project Name	AI-Enhanced Intrusion Detection System
Maximum Marks	4 Marks

Solution Architecture:

The AI-Enhanced Intrusion Detection System is a machine learning-powered cybersecurity web application that classifies network traffic in real time using a Random Forest Classifier trained on NSL-KDD dataset features. The solution architecture below bridges the business problem (detecting and classifying network intrusions) with the technical solution (a Flask-based web application backed by a scikit-learn ML inference pipeline). Its goals are to:

- Find the best tech solution to automatically classify network traffic as Normal or one of four attack types (DoS, Probe, R2L, U2R).
- Describe the structure, components, and data flow of the system to project stakeholders.
- Define the request/response lifecycle, the ML inference layer, and the REST API integration point.
- Provide specifications according to which the solution is built, deployed, and maintained.

Architecture Diagram:

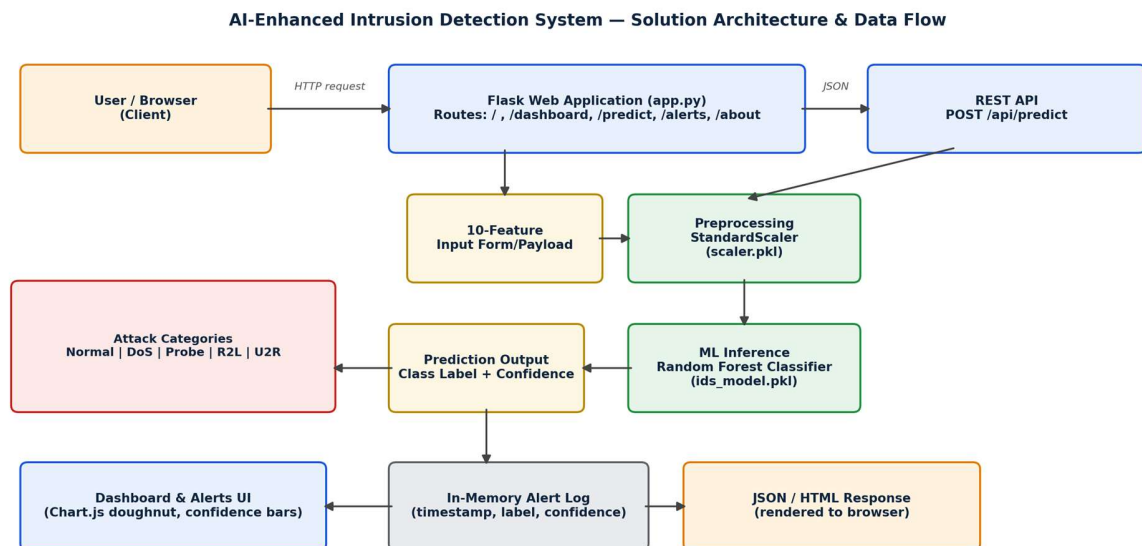


Figure 1: Architecture and data flow of the AI-Enhanced Intrusion Detection System

Component Description:

- User / Browser (Client): End user accesses the web app through routes such as Home, Dashboard, Predict, and Alerts.
- Flask Web Application (app.py): Core MVC-style controller that routes incoming HTTP requests to the correct handler.
- REST API (POST /api/predict): Exposes the same prediction capability programmatically for external system integration, accepting a JSON payload of 10 features.
- Preprocessing (StandardScaler, scaler.pkl): Normalizes the 10 input features to zero mean and unit variance before inference.
- ML Inference (Random Forest Classifier, ids_model.pkl): An ensemble of 100 decision trees trained on the NSL-KDD dataset; returns a class label with a confidence/probability score.
- Prediction Output: The classified label (Normal, DoS, Probe, R2L, or U2R) along with its confidence score.
- In-Memory Alert Log: Stores every prediction with a timestamp, label, and confidence score for auditing.
- Dashboard & Alerts UI: Bootstrap 5 and Chart.js based views that visualize live statistics (doughnut chart) and a full alert history with confidence bars.
- JSON / HTML Response: The final result rendered back to the browser or returned as JSON to API consumers.